Design Component Report

Human-Computer Interaction (HCI)

Assigned Topic: Workplace Interactions & Wellbeing

Course Code: CSE 4451
Section: B

Submitted By

Adiba Habiba	(0112230389)
Rownak Jahan	(011183048)
Faria Benta Alamgir	(0112230202)
Md. Mashrafe Hridoy	(0112230197)
Md. Noman Hossain	(011222159)
Safrin Sanzida Islam	(011212124)

Submitted To

Novia Nurain
Department of Computer Science and Engineering
United International University

Date of Submission

27 January 2026

Contents

1 Introduction **1**

2 Design Motivation and Research Focus **1**

3 Supported Tasks **1**
3.1 Start-of-Day Reflection 2
3.2 Optional In-Day Support 2
3.3 End-of-Day Narrative Reflection 2

4 Low-Fidelity Prototyping **2**

5 Peer Feedback on Low-Fidelity Prototype **3**

6 High-Fidelity Digital Prototype **3**

7 Team Responsibilities **4**

A Additional Design Materials **4**

1 Introduction

This report presents the design component of the *Narrative Weaver* project, following the formative study that explored workplace interactions and well-being in hybrid and organizational work settings. The formative component revealed that employees frequently experience stress due to heavy workload, long working hours, unclear task requirements, and limited opportunities to reflect on the meaning of their work. Participants also expressed mixed attitudes toward AI-supported systems, particularly regarding issues of trust, autonomy, and perceived monitoring. Building on these insights, the design component focuses on translating formative findings into a digital system concept that supports reflection and wellbeing without emphasizing performance evaluation. The objective of this stage is to explore interaction flows and task support through low- and high-fidelity prototyping.

2 Design Motivation and Research Focus

The design of Narrative Weaver is grounded in the research questions established during the formative study, which sought to understand how employees make sense of their work, what contributes to feelings of stress or meaning, and how AI-supported systems might support wellbeing without becoming intrusive. Formative findings indicated that participants felt more positive about their work when they could clearly understand its impact, while repetitive or poorly structured tasks contributed to frustration and mental fatigue. Additionally, participants emphasized the importance of autonomy, noting that wellbeing support should be optional and adaptable to individual preferences. Concerns about monitoring and misuse of AI further shaped the design focus toward transparency, minimal data use, and user control.

3 Supported Tasks

Based on the formative findings, the design supports three core real-world tasks aligned with employees' daily work routines.

3.1 Start-of-Day Reflection

The first task enables employees to reflect briefly at the beginning of the workday by optionally sharing their emotional state and expected workload. This task was motivated by participants' descriptions of fluctuating stress levels depending on workload and deadlines. Successful completion occurs when users are able to provide or skip this reflection voluntarily, reinforcing autonomy and avoiding obligation.

3.2 Optional In-Day Support

The second task provides lightweight, optional support during the workday, such as gentle reminders or reflective prompts. This task addresses formative insights that excessive interruptions increase stress, especially during busy periods. Successful completion is defined by the system adapting its behavior based on user interaction, including dismissal or opt-out, without interrupting focused work.

3.3 End-of-Day Narrative Reflection

The third task supports end-of-day reflection by generating a short narrative summary of the day's activities and contributions. This task was informed by participants' descriptions of feeling satisfied after completing meaningful work or seeing outcomes. Successful completion involves delivering a supportive, non-evaluative narrative that helps users make sense of their work experience.

4 Low-Fidelity Prototyping

Low-fidelity prototyping was conducted using paper sketches to explore interaction flow, system tone, and user control mechanisms. Paper sketches allowed the team to rapidly explore alternative designs for each task and facilitated discussion around when and how the system should appear. The sketches focused on maintaining minimal interaction, clear opt-out options, and a supportive tone rather than detailed interface elements.

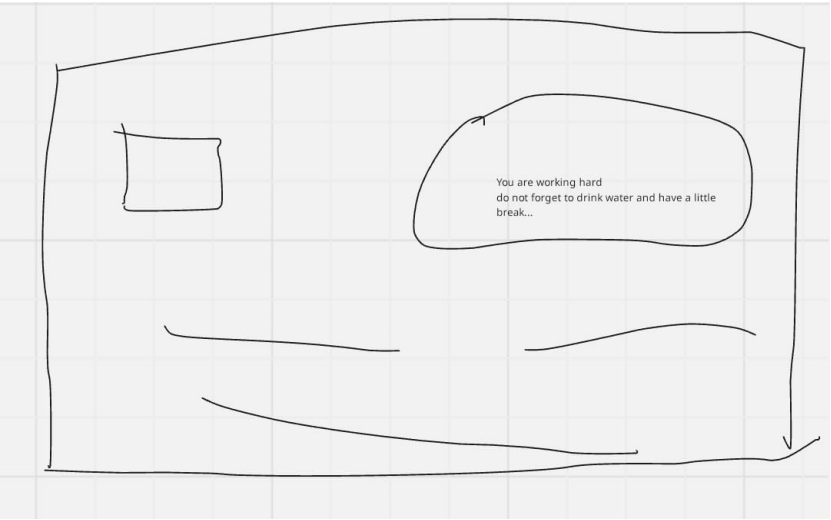


Figure 1: Low-fidelity prototype of an optional midday wellbeing reminder supporting breaks and hydration without interrupting work.

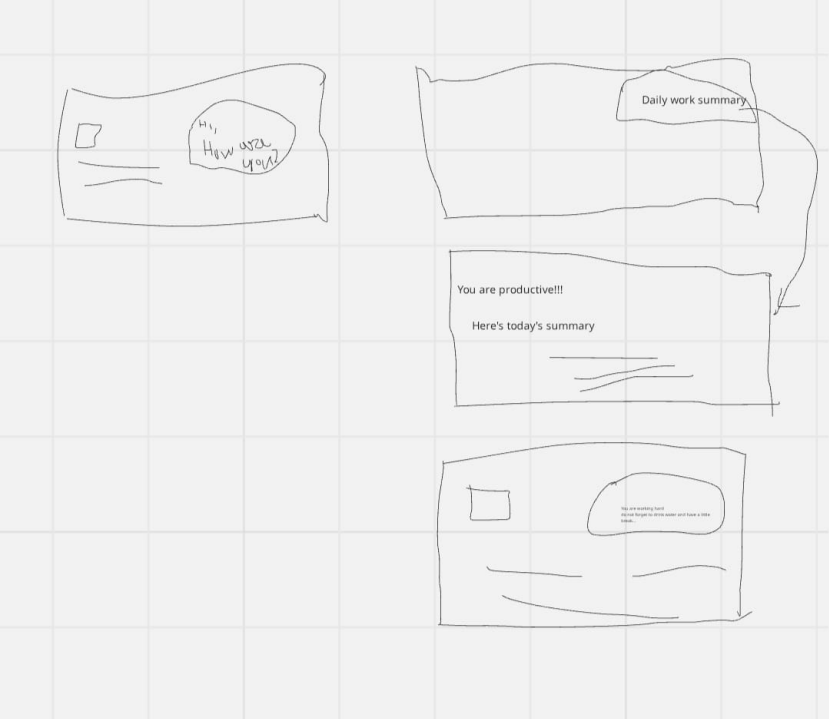


Figure 2: Low-fidelity prototype of codesign

5 Peer Feedback on Low-Fidelity Prototype

Peer evaluation of the paper prototypes highlighted several important considerations. Reviewers emphasized that frequent prompts could feel intrusive, particularly on days with heavy workload. Feedback also reinforced the importance of clear opt-out controls and transparency regarding system behavior. These insights aligned with formative findings and informed the refinement of interaction frequency and messaging tone in the high-fidelity prototype.

6 High-Fidelity Digital Prototype

Informed by peer feedback, the design was translated into a high-fidelity digital prototype using a wireframing and mockup tool. The high-fidelity prototype clarifies interaction sequences while preserving the reflective and non-intrusive nature of the low-fidelity sketches. Changes from the paper prototype include simplified screen layouts, reduced visual emphasis on notifications, clearer user control options, and revised narrative language that avoids performance evaluation.

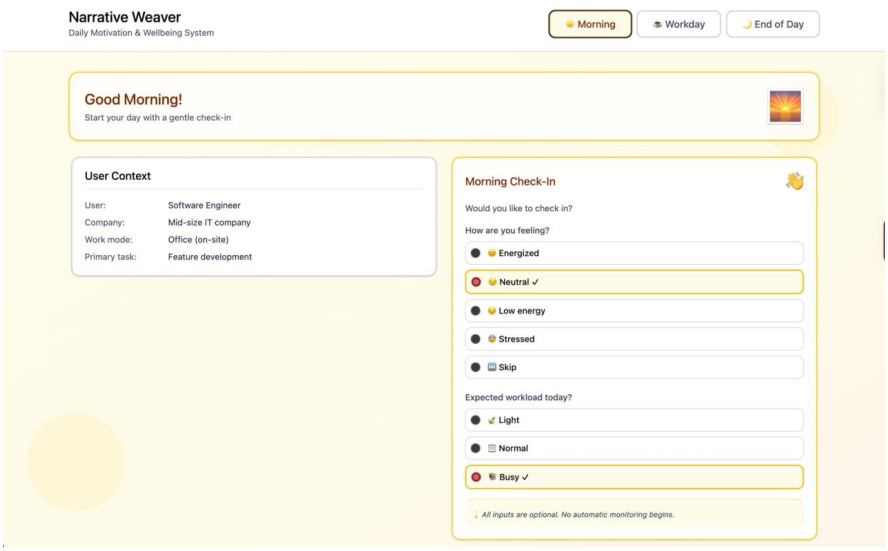


Figure 3: High-fidelity prototype screen supporting start-of-day reflection

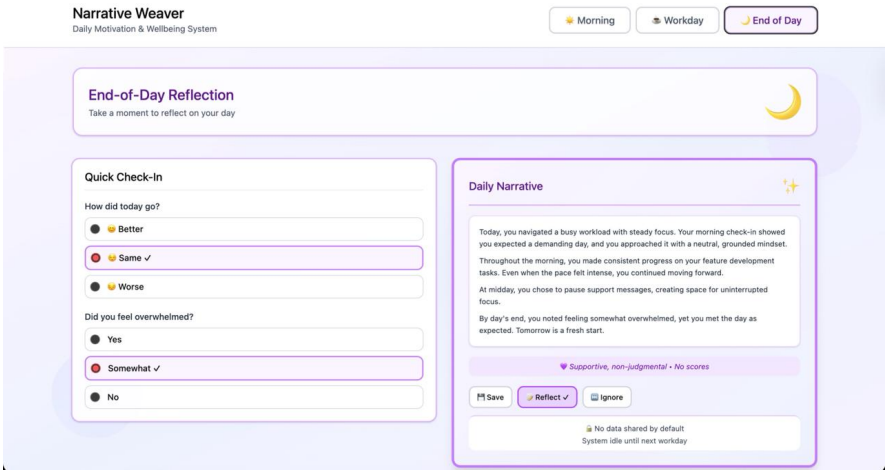


Figure 4: High-fidelity prototype screen supporting End-of-day reflection

7 Team Responsibilities

The design component was carried out collaboratively by all team members. Adiba Habiba coordinated the overall project and prepared the report. Md. Noman Hossain developed the high-fidelity digital prototype. Rownok Jahan contributed to paper prototyping and design feedback. Fariha Benta Alamgir supported task definition and analysis. Md. Mashrafee Hridoy assisted with synthesizing design insights, and Safrin Sanzida Islam contributed to evaluation planning.